

Who Is Shown to Whom?

Sorting, Screening, and Exposure Design

on a Two-Sided Matching Platform*

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Abstract

Two-sided platforms combine decentralized search with recommendations, so recommendation design affects both who meets and how users subsequently search. I study a Japanese medical-staffing platform where doctors apply to short-term posts and platform agents recommend doctors to institutions. Using route-identified contacts and bilateral acceptance decisions, I estimate an equilibrium model of contact, acceptance, and the value of remaining unmatched. The counterfactuals yield three design implications. First, platforms should retain doctor search alongside algorithmic recommendations. Second, a long-run objective that accounts for how continuation values adjust when the recommendation policy changes performs robustly across the platform’s multiple objectives. Third, increasing recommendation volume alone produces modest gains; reallocating recommendations across doctors and posts is substantially more consequential.

JEL Classification Codes: D47; D83; C78; J64; C61; C63; L86.

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1 Introduction

Two-sided matching platforms rarely choose matches directly. Instead, they shape who can match by bringing selected pairs into contact, while participants also search for partners on their own. A recommendation policy therefore operates alongside decentralized search rather

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than in isolation. Because both sides can reject after contact and participants can redirect their own search, a rule that creates more matches immediately need not create the most value over time or distribute matching opportunities broadly across the market. The platform consequently faces three linked choices: whether recommendations should complement or replace participants' own search, whether the algorithm should prioritize immediate match gains, longer-run value, or broad access across the market, and whether to make more recommendations or reallocate those already being made. This paper asks how a platform should make these choices when participants can respond by changing whom they contact on their own.

I answer this question using data from a Japanese platform where doctors apply to short-term posts and platform agents recommend doctors to institutions. The data identify the route initiating each contact and, when defined, both sides' decisions. I estimate an equilibrium model linking route-specific contact, bilateral acceptance, and the value of remaining unmatched, then compare recommendation policies as doctor search adjusts. Recommendations work best as a complement to doctor search. Relative to the current hybrid, the best-found long-run redesign raises model-implied long-run value by 2.11 percent when search remains available, while replacing search lowers it by 9.36 percent even under the best-found long-run rule. Doctors' ability to redirect contacts keeps long-run performance high across objectives. Among the policies evaluated, the long-run rule is therefore a natural default: it attains the highest best-found long-run value, preserves 98.34 percent of the maximum immediate-match objective, and distributes post-side outcomes more evenly than pure short-run targeting.

Section 2 describes the Japanese two-sided labor platform and the records used in the analysis. A contact occurs either when a doctor searches and applies for a post or when a platform agent recommends a doctor for a post, and the data record both the initiating route and the subsequent decision of each side when that decision is defined. The analysis focuses on 1,132 doctors and 2,412 posts scheduled in the Kanto region in December 2024, constructs predetermined doctor–facility histories and a doctor activity proxy from earlier platform use, and documents a sharp reversal across routes: following doctor-initiated search, doctor and post acceptance rates are 98.0 and 41.0 percent, whereas following agent-initiated screening they are 28.3 and 98.4 percent.

Section 3 develops a continuous-time two-sided search model that preserves these two contact routes. Doctors operate one or more interchangeable slots, each doctor–post pair has route-specific Poisson contact rates, and a match forms only if both sides accept after contact. The initiating side's continuation value affects its contact rate, while contact rates and bilateral acceptance jointly determine continuation values. Stationary renewal closes this feedback and provides the equilibrium system used in estimation and the policy exercises.

Section 4 parameterizes two route-specific contact indices and two side-specific acceptance indices using doctor, post, and pair characteristics, including five predetermined measures of

prior doctor–facility relationships. The likelihood combines route-specific Poisson approach counts with doctor- and post-side acceptance decisions, and the parameters and continuation values are estimated jointly subject to the stationary-equilibrium restriction. The fitted model matches the aggregate number of approaches in each route and preserves the qualitative reversal in acceptance across routes, although it understates all four acceptance rates. Distance reduces fitted contact and acceptance indices in every case; most compatibility and prior-relationship variables move in the expected direction, although some conditional signs remain counterintuitive.

Section 5 first compares recommendation policies that leave doctor search available with policies that replace it with contacts assigned entirely by the platform. Within each class, it compares an even allocation, a rule maximizing immediate expected match gains, and a rule that accounts for continuation values after equilibrium adjustment. The results favor retaining doctor search: when doctors can adjust their own contacts, long-run value remains high across recommendation rules, whereas removing doctor search makes both value and post-side access much more sensitive to what the algorithm targets. Conditional on retaining search, the best-found long-run rule preserves 98.34 percent of the maximum short-run objective, produces more even post-side outcomes than the short-run rule, and redirects recommendations toward shorter, closer, and more compatible posts.

Section 5 then separates three sequential policy margins: increasing the total number of recommendations, equalizing expected recommendation totals across doctors, and redesigning which posts receive them. With doctor search retained, doubling the fitted current recommendation profile raises aggregate continuation value by 2.23 percent; holding the doubled volume fixed, equalizing recommendation totals adds 10.25 percent, and holding those equal totals fixed, redesigning their allocation across posts adds a further 9.58 percent. Thus, the larger gains come from changing the distribution of recommendations rather than increasing their volume alone. Because the calculation does not subtract the platform’s recommendation costs, it does not identify the optimal number of recommendations. In a market-expansion benchmark that adds stock to top-decile doctor or post types, reproducing the redesign gain requires 9.31 percent more doctor-slot stock when post stock is fixed, 26.80 percent more post stock when doctor stock is fixed, or 1.42 percent more doctor-slot stock after post stock first increases by 20 percent. These equivalences imply that algorithmic redesign can partly substitute for one-sided market expansion and that doctor and post capacity are complementary.

Related literature. The paper first contributes to the frictional matching literature on sorting and screening (Eeckhout and Kircher, 2010; Moen, 1997; Burdett and Coles, 1997; Shimer and Smith, 2000; Chade, Eeckhout and Smith, 2017) by making the contrast between doctor-initiated search and platform-initiated screening empirically operational. The same platform

contains both routes, the data identify which route generated each contact, and the model evaluates alternative channel architectures and allocation rules for a given number of platform-assigned contacts.

Second, the paper relates to empirical work on online matching platforms (Hitsch, Hortacısu and Ariely, 2010; Fradkin, 2017; Gong, 2016; Agarwal, 2015; Galichon, Kominers and Weber, 2019). Its distinctive feature is observing the initiating route and the subsequent doctor- and post-side decisions when those decisions are defined, rather than only realized matches.

Third, the paper connects to work on recommendations and steering in labor and platform markets (Horton, 2017; Belot, Kircher and Muller, 2019; Le Barbanchon, Hensvik and Rathelot, 2023; Chen and Tsai, 2024), which estimates the effects of providing or changing recommendations. Here recommendations are one route inside a structural two-route system, so the counterfactuals compare alternative channel architectures and allocation rules on a common footing.

Finally, the paper contributes to algorithmic and operations work on search design, multichannel traffic, fairness, and two-sided recommendation (Immorlica et al., 2023; Manshadi et al., 2025; Shi, 2023, 2025; Chen, Hsieh and Lin, 2023; Sürer, Burke and Malthouse, 2018; Biega, Gummadi and Weikum, 2018): the recommendation rule is evaluated through equilibrium continuation values under finite doctor capacity rather than through immediate conversion or ranking metrics alone.

2 Setting and Data

2.1 Platform and Exposure Channels

The empirical setting is a Japanese two-sided labor platform operated by Medical Principle Co., Ltd. The platform matches licensed doctors with short-term “spot” positions at hospitals and clinics. Participating doctors typically hold regular positions elsewhere. Hospitals and clinics post temporary openings, primarily for health checkups, ward coverage, and overnight on-call shifts. The platform records the contacts and applications associated with each opening, as well as the subsequent decisions by both sides that lead to a match.

Contact between a doctor and a post can arise through either of two routes. A doctor may search the platform and apply to a post; I refer to the resulting recorded contact as *doctor-initiated search*, or the *sorting route*. Alternatively, a platform agent may identify a doctor and recommend that doctor for a post; I refer to this as *agent-initiated screening*, or the *screening route*. Medical institutions do not directly search the platform’s doctor pool. Under the platform’s current operation, doctor-initiated search therefore reflects information and initiative on the doctor side, whereas agent-initiated screening is the route that the platform

can directly redesign.¹

I use *approach* to refer to either type of recorded contact. An approach is distinct from a listing view: it brings a particular doctor–post pair into contact and initiates an exchange of detailed information through the intermediary, often in the form of an interview. Both sides then decide whether to proceed, and a match forms only if both sides accept. A doctor who initiated an approach may still decline after learning the details, and either side may reject a recommendation initiated by an agent.

2.2 Market and Measurement

The online job-matching platform operates continuously, so the empirical analysis must isolate a specific market from the ongoing flow of doctors and posts. This subsection first explains how I define the estimation cohort. It then describes how I construct bilateral acceptance decisions and predetermined measures of prior platform activity within that cohort, before presenting its main descriptive patterns.

Estimation cohort. I define the estimation cohort as posts scheduled in the Kanto region in December 2024 and all doctors appearing in at least one approach to those posts. Because an application may be submitted before a post’s scheduled work date, I retain approaches within each selected post’s episode-specific half-open observation window rather than impose a December cutoff on approach dates. The final cohort contains 1,132 doctors and 2,412 posts. Here, for the purpose of defining the active market, I classify doctors who neither approach a selected post nor receive an agent-initiated approach to one as *inactive* in the December 2024 spot-job market. In other words, these doctors generate no recorded indication that they are actively considering a post in this cohort. This is an operational definition based on observed platform activity; it does not rule out unrecorded browsing or interest in other spot jobs.

The data also contain basic characteristics of both sides. Doctor characteristics include age, experience, specialty, location, and stated preferences. Post characteristics include location, task content, desired specialty, schedule, working hours, and compensation.

Route and decision construction. The data record the initiating route and subsequent status of each approach, from which I construct separate doctor- and post-side acceptance indicators. Repeated in-window approaches remain distinct events in the Poisson exposure counts. For acceptance, a doctor–post pair whose panel decision can be attributed to exactly one current-episode route contributes at most one Bernoulli trial for each side with an observed

¹The platform’s user interface is itself a potential object of design. For example, the platform could remove the search function available to doctors and instead generate all approaches through algorithmic recommendations. Section 5 considers this more substantial change to platform functionality as a counterfactual benchmark.

Table 1. Mapping from Approach Status to Acceptance Indicators

Approach status	Doctor accepts	Post accepts
Contract	1	1
Cancelled After Contract	1	1
Not Hired	1	0
Approach	0	1
Inquiry Handled	0	–

Notes: 1 denotes acceptance, 0 denotes rejection, and – denotes an undefined decision. An undefined decision is treated as missing rather than as a rejection.

decision; both-route and unassignable pairs do not enter the route-specific acceptance denominators. For each approached doctor–post pair, the platform records one of five statuses: *Contract*, *Cancelled After Contract*, *Not Hired*, *Approach*, or *Inquiry Handled*. The mapping from these statuses to binary acceptance indicators, constructed in consultation with the platform’s staff, is reported in Table 1.

Past platform activity. I construct five predetermined variables separately for each doctor–facility pair, using only activity recorded on the platform strictly before the order date of the current post: the number of approaches made by the doctor to the facility during the preceding 365 days; an indicator for any prior contract between the doctor and the facility; the number of days since the doctor’s most recent approach to the facility; the share of the pair’s approaches during the preceding 365 days that were initiated by an agent; and an indicator that no prior relationship between the doctor and the facility is observed. These variables place recorded traces of persistent familiarity in the four empirical indices. They do not eliminate unobserved doctor–facility heterogeneity or latent pair quality.

The formal model also allows doctors to differ in how many posts they can search simultaneously. I interpret this latent quantity as an activity or throughput measure, motivated by the substantial differences in recorded platform contacts among otherwise observably similar licensed doctors. Because this quantity is not directly observed, I proxy it with the maximum, over September–November 2024, of a doctor’s monthly number of recorded approaches across both routes, excluding approaches to the final December cohort and imposing a floor of one slot.

Descriptive patterns. Table 2 summarizes the standard market variables. Table 3 separately reports the five predetermined doctor–facility history variables.

A notable feature of the data is the sharp asymmetry according to who initiates an approach. Among observed decisions following doctor-initiated search, doctors accept 98.0 percent and

posts accept 41.0 percent. Following agent-initiated screening, the corresponding acceptance rates are 28.3 percent for doctors and 98.4 percent for posts. Thus, doctor-initiated search reflects substantial doctor-side preselection, whereas agent-initiated screening reflects substantial post-side preselection.

Controlling for past doctor–facility activity is particularly important when estimating these acceptance patterns. Only 121,065 of the 2,730,384 possible doctor–post pairs, or 4.43 percent, have a nonzero prior doctor–facility history. Among the 3,281 route-attributable doctor–post cells eligible for at least one side’s acceptance likelihood, however, 1,685 cells, or 51.36 percent, have such history. These route-attributable pair cells are therefore much more likely than an average possible doctor–post pair to involve a doctor and facility with a pre-existing relationship. Without these recorded-history variables, differences associated with prior familiarity could be attributed to the initiating route or to contemporaneous pair characteristics.

Table 2. Market Descriptive Statistics

Variable	<i>N</i>	Mean	SD	P25	Median	P75	Nonzero (%)
<i>A. Doctor characteristics</i>							
Approach events per doctor	1,132	3.11	3.91	—	2.00	—	—
Age	1,132	42.74	12.06	33.00	40.00	50.25	100.00
Experience	1,132	16.49	11.77	7.00	13.00	24.00	100.00
Lagged activity/throughput proxy K_i	1,132	3.70	4.91	1.00	2.00	4.00	100.00
<i>B. Post characteristics</i>							
Approach events per post	2,412	1.46	1.32	—	1.00	—	—
Hours	2,412	10.12	8.68	4.00	8.50	12.50	100.00
Advertised pay	2,412	72.55	43.42	44.00	60.50	91.25	100.00
Advertised wage per hour	2,412	9.13	3.91	5.83	10.00	11.43	100.00
<i>C. Doctor–post characteristics</i>							
Distance	2,730,384	61.08	104.81	22.76	34.40	58.32	100.00
Preferred-location match (%)	2,730,384	25.63	43.66	0.00	0.00	100.00	25.63
Preferred-day match (%)	2,730,384	31.49	46.45	0.00	0.00	100.00	31.49
Specialty match (%)	2,730,384	15.84	36.51	0.00	0.00	0.00	15.84
Outpatient-task match (%)	2,730,384	22.72	41.91	0.00	0.00	0.00	22.72
Ward-task match (%)	2,730,384	13.50	34.18	0.00	0.00	0.00	13.50

Notes: Entries are raw sample statistics. Indicator variables are shown in percentage points. Age and experience are in years; hours are hours; distance is in kilometers; advertised pay is in thousand yen; and hourly wage is in thousand yen per hour. Dashes mark unavailable quantiles or nonzero shares. The lagged activity measure is a throughput proxy, not observed physical capacity.

Table 3. Predetermined Doctor–Facility Relationship Variables

Variable	<i>N</i>	Mean	SD	P25	Median	P75	Nonzero (%)
Prior approaches (365 days)	2,730,384	0.11	1.92	0.00	0.00	0.00	2.38
Any prior contract (%)	2,730,384	2.00	13.98	0.00	0.00	0.00	2.00
Days since last approach	2,730,384	19.58	127.11	0.00	0.00	0.00	4.43
Prior agency share (365 days) (%)	2,730,384	0.49	6.55	0.00	0.00	0.00	0.66
No prior relationship (%)	2,730,384	95.57	20.58	100.00	100.00	100.00	95.57

Notes: History is measured strictly before each post’s order date and shown before estimation standardization.

Days since last approach and prior agency share are coded zero when no prior relationship exists; the separate no-prior-relationship indicator records that boundary.

3 Model

I develop a continuous-time variant of the two-sided search model of Adachi (2003). The two sides of the market are doctors and posts. On the doctor side, I call the unit that can form a match a *doctor slot*. A doctor may operate several interchangeable slots, which allows the model to represent differences in how actively doctors search and, in particular, how many spot jobs they consider concurrently. Observed characteristics and preferences are shared across slots belonging to the same doctor, but each slot is treated as an agent in the model. No analogous construction is needed on the post side: because an open post leaves the market once it matches, each post is itself one model agent.

The sequence of events in continuous time is as follows. At any instant, each unmatched doctor slot and each open post is in its own waiting state and has a continuation value that summarizes its future matching opportunities. For each doctor–post pair, the doctor chooses the intensity of doctor-initiated search, while a platform agent acting on behalf of the post chooses the intensity of agent-initiated screening. These pair- and route-specific intensities govern independent Poisson clocks. When one of these clocks rings, an approach occurs. The doctor and post then observe the information revealed by that contact and independently decide whether to proceed. If both accept, a match forms and the matched doctor slot and post exit. If either rejects, the doctor slot remains unmatched and the post remains open, so their contact processes continue.

The remainder of the section follows this order: I first describe approach-intensity choice and arrival, then post-approach acceptance, then continuation values and stationary renewal.

3.1 Agents, Timing, and Approach Decisions

Agents and exposure routes. Time is continuous. Let I be the set of doctors, J the set of posts, and $\mathcal{C} = \{\text{self}, \text{agency}\}$ the two exposure routes. The route *self* denotes a doctor-initiated approach generated by the doctor’s search over listings. The route *agency* denotes an agent-initiated recommendation generated by screening on behalf of the post. Doctor i has $K_i \geq 1$ interchangeable doctor slots. Each post j is one open position and therefore one model agent. Each doctor slot can receive approaches through both routes.

Choosing approach intensities. For each doctor–post pair, the initiator of each route chooses the instantaneous rate at which that route generates an approach. On the self route, a higher rate represents more search attention by the doctor toward the post and therefore a higher arrival hazard of an application. On the agency route, it represents more screening attention directed toward the pair and therefore a higher arrival hazard of a recommendation. I represent this decision as a pair- and route-specific choice of a Poisson arrival rate. The rate

determines both the expected number of approaches per unit of time and the instantaneous hazard that an approach occurs. The initiator weighs the gross value of generating an approach against the continuation value of waiting, while moving the rate away from a route-specific baseline incurs a convex adjustment penalty.

Let s_{ij}^c denote the route-specific gross value that the initiator assigns to generating one additional approach between doctor i and post j . This is a pre-contact value index. It is distinct from the match payoff that the two sides observe after an approach occurs and from the total-user-value objective used for policy evaluation.

The relevant opportunity cost in the intensity choice is the continuation value of the initiating side:

$$\omega_{ij}^c(\alpha) = \begin{cases} \alpha_i^D, & c = \text{self}, \\ \alpha_j^P, & c = \text{agency}, \end{cases}$$

where α_i^D is the value of keeping one slot of doctor i available for future matching opportunities, and α_j^P is the value of keeping post j open. At this stage, these are candidate continuation values. The stationary equilibrium defined below requires them to coincide with the values generated by the resulting approach-arrival and acceptance processes.

For a candidate Poisson rate $\lambda > 0$ and a route-specific reference rate $\bar{\lambda}^c > 0$, define

$$D_{\text{KL}}(\lambda \| \bar{\lambda}^c) = \lambda \log\left(\frac{\lambda}{\bar{\lambda}^c}\right) - \lambda + \bar{\lambda}^c.$$

This expression is the relative-entropy rate of the Poisson process with intensity λ relative to the reference process with intensity $\bar{\lambda}^c$. It is nonnegative, equals zero at the reference rate, and is strictly convex in λ . I use it as an auxiliary convex penalty for moving the search or screening process away from its baseline, rather than interpreting it as a separately identified structural effort cost.²

I measure the two gross approach-value indices in units that normalize the intensity-adjustment scales to $\zeta^{\text{self}} = \zeta^{\text{agency}} = 1$. The initiator's auxiliary choice problem in the current market is therefore

$$\max_{\lambda > 0} \lambda(s_{ij}^c - \omega_{ij}^c(\alpha)) - D_{\text{KL}}(\lambda \| \bar{\lambda}^c). \quad (1)$$

The first term is the flow value of generating approaches, net of the initiator's waiting-value opportunity cost. The second term penalizes moving the approach rate away from its reference level. Its marginal cost is $\log(\lambda/\bar{\lambda}^c)$, so the objective is strictly concave and the unique conditional choice satisfies

$$\log \lambda_{ij}^c(\alpha) = \log \bar{\lambda}^c + s_{ij}^c - \omega_{ij}^c(\alpha). \quad (2)$$

²Relative-entropy penalties for moving a controlled stochastic law away from a passive reference law are used in KL-control formulations; see Todorov (2006). Wang et al. (2017) applies this idea directly to the control of point-process intensities. Here the device only rationalizes the functional form of the approach-rate equation: the penalty itself does not enter continuation values or the policy objective.

A positive net value raises the rate above that baseline, while a negative net value lowers it. On the self route, the doctor makes this choice. On the agency route, I use the same rule for the platform agent screening on behalf of the post.

Given the chosen rates, approaches between each slot of doctor i and post j arrive through two independent Poisson processes, with rates $\lambda_{ij}^{\text{self}}$ and $\lambda_{ij}^{\text{agency}}$. The two clocks almost surely do not ring simultaneously, so their rates add without an overlap correction. Across doctor i 's K_i slots, the aggregate route- c approach rate to post j is $K_i \lambda_{ij}^c$.

Acceptance after an approach. An approach is not a match. It is an event at which the two sides observe their match-specific information and then make bilateral acceptance decisions. Conditional on route c , their one-time payoffs are

$$U_{ij}^c = u_{ijc}^D + \varepsilon_{ijc}^D, \quad (3)$$

$$V_{ji}^c = u_{jic}^P + \varepsilon_{jic}^P. \quad (4)$$

The shocks are independent logistic random variables with scales $\zeta^D, \zeta^P > 0$, drawn independently across sides and exposure events. Utility is nontransferable, and the pair matches if and only if both sides accept. The systematic terms u_{ijc}^D and u_{jic}^P may depend on the route.

The acceptance rules are

$$a_{ijc}^D = \mathbf{1}\{U_{ij}^c \geq \alpha_i^D\}, \quad a_{jic}^P = \mathbf{1}\{V_{ji}^c \geq \alpha_j^P\}.$$

Write the resulting route-specific acceptance probabilities as

$$P_{ijc}^D(\alpha) = \frac{1}{1 + \exp\{(\alpha_i^D - u_{ijc}^D)/\zeta^D\}}, \quad P_{jic}^P(\alpha) = \frac{1}{1 + \exp\{(\alpha_j^P - u_{jic}^P)/\zeta^P\}},$$

where u_{ijc}^D and u_{jic}^P denote the systematic parts of (3) and (4). Bilateral acceptance on route c occurs with probability $p_{ijc}(\alpha) = P_{ijc}^D(\alpha)P_{jic}^P(\alpha)$.

3.2 Stationary Renewal and Continuation Values

Let $\delta_i^a \geq 0$ be doctor i 's exogenous abandonment hazard, $w_j \geq 0$ post j 's withdrawal hazard, and $r > 0$ the discount rate. The following benchmark turns the two acceptance thresholds into stationary continuation values.

Assumption 1 (Stationary renewal).

- (i) (Time-invariant primitives) *Gross approach values, baseline rates, payoff distributions, stocks, exit hazards, and discounting do not vary over time.*
- (ii) (Renewal) *When a doctor slot exits through a match or abandonment, or a post exits through a match or withdrawal, it is immediately renewed by a unit with the same modelled type. Hence the type composition and contact environment remain fixed.*

(iii) (Absorbing payoffs) *A matched unit receives its one-time payoff and exits; exogenous exit yields zero. The renewed unit starts a new search problem with independent future shocks.*

I model the market as stationary. In addition to leaving after a match, a doctor slot may become inactive and an open post may be withdrawn before matching. Either event removes that unit from the active market for reasons outside the model. Under stationary renewal, every exiting unit is immediately replaced by another unit of the same modeled type. Market stocks and the marginal type distributions on both sides therefore remain fixed. This is an analytical approximation, not a claim that the platform literally replaces each departing unit with an identical clone.³ Because the empirical market is calibrated to a single monthly cohort, I hold the marginal type distributions fixed when comparing stationary policies.

For route c , define the expected own-side gain above continued waiting,

$$q_{ijc}^D(\alpha) = \mathbb{E}[(U_{ij}^c - \alpha_i^D)_+], \quad q_{ijc}^P(\alpha) = \mathbb{E}[(V_{ji}^c - \alpha_j^P)_+],$$

where $x_+ = \max\{x, 0\}$. Because the two payoff shocks are independent, the expected gains from an exposure, taking the other side's acceptance into account, are

$$\psi_{ijc}^D(\alpha) = P_{jic}^P(\alpha) q_{ijc}^D(\alpha), \quad \psi_{ijc}^P(\alpha) = P_{ijc}^D(\alpha) q_{ijc}^P(\alpha).$$

The route matters both through its contact rate and through the systematic post-approach payoffs inside P , q , and ψ .

Balancing the exposure gains against discounting and exogenous exit gives the route-aware continuation-value system

$$\begin{aligned} \alpha_i^D &= \Phi_{i,K}^D(\alpha; \lambda) := \frac{\sum_{j \in J} \sum_{c \in \mathcal{C}} \lambda_{ij}^c \psi_{ijc}^D(\alpha)}{r + \delta_i^a}, \\ \alpha_j^P &= \Phi_{j,K}^P(\alpha; \lambda) := \frac{\sum_{i \in I} \sum_{c \in \mathcal{C}} K_i \lambda_{ij}^c \psi_{ijc}^P(\alpha)}{r + w_j}. \end{aligned} \tag{5}$$

The difference between the two equations reflects units: λ_{ij}^c is the contact rate for one doctor slot, whereas $K_i \lambda_{ij}^c$ is the aggregate flow from doctor i to post j .

Let $\Phi_K(\alpha; \lambda)$ stack the right-hand sides of (5), and define

$$\mathcal{Z}(\lambda^{\text{self}}, \lambda^{\text{agency}}, K) = \{\alpha \in \mathbb{R}_{\geq 0}^{|I|+|J|} : \alpha = \Phi_K(\alpha; \lambda)\}.$$

Definition 1 (Fixed-rate stationary-renewal equilibrium). *Fix $(\lambda^{\text{self}}, \lambda^{\text{agency}}, K)$. A fixed-rate stationary-renewal equilibrium is a pair (a, α) such that each side uses its route-specific threshold rule after every exposure and α equals the expected discounted value of waiting generated by those rules under Assumption 1.*

³Richer steady-state search models instead allow entry and matching jointly to determine the composition of the unmatched stocks; see, for example, Lauermaun and Nöldeke (2025). Allowing the type distributions to evolve as units enter and leave the market would be substantially less tractable at the scale of this application.

Proposition 1 (Fixed-rate characterisation and existence). *For nonnegative finite route rates and finite I, J , the fixed-rate stationary-renewal equilibria are exactly the threshold rules generated by $\alpha \in \mathcal{Z}(\lambda^{\text{self}}, \lambda^{\text{agency}}, K)$, and this set is nonempty.*

Proof. See Appendix A.3. □

Proposition 1 characterizes equilibrium conditional on fixed route-specific approach rates. In the current market, however, these rates are themselves chosen and therefore depend on the continuation values. Let $\Gamma(\alpha; s)$ stack the behavioral rate rules in equation (2). The current-market equilibrium combines rate optimality with value consistency:

$$\lambda = \Gamma(\alpha; s), \quad \alpha = \Phi_K(\alpha; \lambda). \quad (6)$$

A behavioral stationary-renewal equilibrium is a triple (λ, a, α) that satisfies equation (6) and the route-specific threshold rules. Because the behavioral rates are continuous, nonnegative, and bounded above on the nonnegative orthant by their values at zero, the same compact-box argument extends fixed-rate existence to this joint system; Appendix A.3 gives the argument.

4 Empirical Strategy and Results

4.1 Parameterization and Likelihood

The model leaves the gross approach values s_{ij}^c and the systematic post-approach payoffs u_{ijc}^D and u_{jic}^P unrestricted. To connect these primitives to the observed approach counts and bilateral decisions, I parameterize them using doctor characteristics, post characteristics, and variables defined for each doctor–post pair. I then impose the equilibrium consistency condition from Section 3.2 when evaluating the likelihood.

Let $c \in \{\text{self}, \text{agency}\}$ denote doctor-initiated search and agent-initiated screening. Let Z_{ij}^D and Z_{ij}^P denote the two design rows used by the implementation. Each collects three groups of observed covariates: doctor characteristics, post characteristics, and variables defined for the doctor–post pair. The third group includes distance, indicators for preference, schedule, specialty, and task compatibility, and the five predetermined doctor–facility history measures. The first column of each design row is identically one. Its coefficient is the only intercept in the corresponding index.

Gross approach values and contact rates. Each doctor–post pair has two route-specific contact rates: one for doctor-initiated search and one for agent-initiated screening. These rates determine the expected number of approaches observed through each route. Because the reference rate $\bar{\lambda}^c$ in equation (2) is constant within a route, I absorb $\log \bar{\lambda}^c$ into a route-specific

constant. I write the resulting empirical parameterization as

$$\begin{aligned}s_{ij}^{\text{self}} + \log \bar{\lambda}^{\text{self}} &= (Z_{ij}^D)' \beta_{\text{self}}, \\ s_{ij}^{\text{agency}} + \log \bar{\lambda}^{\text{agency}} &= (Z_{ij}^P)' \beta_{\text{agency}}.\end{aligned}$$

Substituting these expressions into the model’s intensity rule gives

$$\begin{aligned}\log \lambda_{ij}^{\text{self}} &= (Z_{ij}^D)' \beta_{\text{self}} - \alpha_i^D, \\ \log \lambda_{ij}^{\text{agency}} &= (Z_{ij}^P)' \beta_{\text{agency}} - \alpha_j^P.\end{aligned}$$

A doctor’s continuation value reduces doctor-initiated search, while a post’s continuation value reduces agent-initiated screening. We normalize $\zeta^{\text{self}} = \zeta^{\text{agency}} = 1$; there is no separately estimated exposure-scale parameter.

Post-approach payoffs and acceptance. The same three groups of covariates enter the two post-approach payoff indices, but their coefficients differ by side and from the coefficients in the contact-rate equations. I parameterize the systematic payoffs in equations (3)–(4) as

$$\begin{aligned}u_{ijc}^D &= (Z_{ij}^D)' \beta_D + \chi_D \mathbf{1}\{c = \text{self}\}, \\ u_{jic}^P &= (Z_{ij}^P)' \beta_P + \chi_P \mathbf{1}\{c = \text{agency}\}.\end{aligned}$$

Because payoff coefficients and logit scales are not separately identified, I normalize $\zeta^D = \zeta^P = 1$. The resulting acceptance probabilities are

$$\begin{aligned}\Pr(a_{ijc}^D = 1) &= \Lambda\left((Z_{ij}^D)' \beta_D - \alpha_i^D + \chi_D \mathbf{1}\{c = \text{self}\}\right), \\ \Pr(a_{jic}^P = 1) &= \Lambda\left((Z_{ij}^P)' \beta_P - \alpha_j^P + \chi_P \mathbf{1}\{c = \text{agency}\}\right).\end{aligned}$$

The two χ coefficients are route-specific differences in acceptance intercepts. For doctors, agent-initiated screening is the reference route, so χ_D is the change in the doctor acceptance index when the doctor initiates the contact. For posts, doctor-initiated search is the reference route, so χ_P is the change in the post acceptance index when a platform agent initiates the contact.

Observed likelihood and equilibrium restriction. The likelihood is defined over the doctor–post pairs constructed in Section 2.2. For each selected post episode in the monthly market, the observed route-specific approach count satisfies

$$n_{ij}^c \sim \text{Poisson}\left(K_i T_j \lambda_{ij}^c\right),$$

where λ_{ij}^c is the monthly contact rate for one doctor slot and T_j is the post episode’s observed duration in months. Doctor i has K_i such slots, so $K_i T_j \lambda_{ij}^c$ is the exposure-adjusted expected approach count for the doctor–post pair. The predetermined activity proxy fixes K_i before estimation and it enters with no estimated coefficient.

Write $\mathcal{C} = \{\text{self}, \text{agency}\}$ and let $\mathcal{M}$ denote the set of doctor–post pairs in the estimation market. For route c , let $\mathcal{A}_c^D$ and $\mathcal{A}_c^P$ denote the subsets with an observed doctor- and post-side acceptance decision, respectively. Set $m_{ij}^c = K_i T_j \lambda_{ij}^c$, $P_{ijc}^D = \Pr(a_{ijc}^D = 1)$, and $P_{jic}^P = \Pr(a_{jic}^P = 1)$. Suppressing the dependence of the rates and acceptance probabilities on (θ, α) , the likelihood is

$$\begin{aligned} L(\theta, \alpha) = & \prod_{c \in \mathcal{C}} \prod_{(i,j) \in \mathcal{M}} \frac{\exp(-m_{ij}^c) (m_{ij}^c)^{n_{ij}^c}}{(n_{ij}^c)!} \\ & \times \prod_{c \in \mathcal{C}} \prod_{(i,j) \in \mathcal{A}_c^D} (P_{ijc}^D)^{a_{ijc}^D} (1 - P_{ijc}^D)^{1-a_{ijc}^D} \\ & \times \prod_{c \in \mathcal{C}} \prod_{(i,j) \in \mathcal{A}_c^P} (P_{jic}^P)^{a_{jic}^P} (1 - P_{jic}^P)^{1-a_{jic}^P}. \end{aligned} \quad (7)$$

Because $\mathcal{C}$ contains two routes, equation (7) has two Poisson count components and four route-by-side Bernoulli components. The route-specific constants inside λ_{ij}^c remain elements of θ and are estimated jointly rather than profiled out.

4.2 Identification and Estimation Procedure

The contact records and the decisions made after contact discipline different parts of the model jointly with the continuation values. Variation in doctor-initiated and agent-initiated approach counts across pairs informs the two contact-rate indices. Conditional on a route-attributable approach, variation in doctor and post decisions across pairs and routes informs the two acceptance indices and the two initiator-specific acceptance intercept differences. These sources of variation are interpreted conditional on the included covariates and maintained equilibrium branch; they are not a formal point-identification result for the full constrained system.

A remaining identification concern is pair-specific information that doctors or platform agents observe but the researcher does not. For example, a doctor may apply to a hospital, or an agent may recommend that doctor, because the doctor has worked successfully with the facility before. The same familiarity can also make both sides more likely to accept after the approach. If it is omitted, the estimated coefficients on distance, recorded match indicators, or the identity of the initiating side may partly capture relationship-driven contact and acceptance. I therefore include the five predetermined doctor–facility history measures in every contact-rate and acceptance index. These measures record prior approaches, prior contracting, recency, the route used in earlier contacts, and the absence of any recorded prior relationship. They place the most direct observable traces of persistent doctor–facility familiarity in both stages of the model.

Another limitation arises because existence of the stationary-renewal fixed point does not imply that its solution is unique. If the same structural parameters support several continuation-value solutions, the mapping from parameters to observed outcomes is not single valued, and

the data alone need not reveal which equilibrium generated the market. As in many dynamic structural models, the present analysis does not resolve this general multiple-equilibrium identification problem. The estimates therefore condition on the equilibrium branch implied by the documented initialization.

The monthly doctor-abandonment hazard, post-withdrawal hazard, and discount rate are calibrated to 0.05, 0.10, and $-\log(0.99)$, respectively. I formulate the numerical target as joint maximization of $\ell(\theta, \alpha) = \log L(\theta, \alpha)$ over the parameters and continuation values, subject to equation (6). Appendix Section B.3 describes the MPEC problem, solver implementation, and convergence diagnostics.

4.3 Estimates and Channel Fit

The MPEC estimate matches the aggregate number of approaches in each route: the observed and fitted counts are 2,503 for doctor-initiated sorting and 1,013 for agent-initiated screening. Acceptance fit is less exact, although the model preserves the qualitative reversal across routes. Doctor acceptance is 97.98 percent in the data and 87.12 percent in the model after sorting, and 28.35 versus 8.50 percent after screening. Post acceptance is 40.98 versus 27.37 percent after sorting and 98.43 versus 94.52 percent after screening.

Table 4 puts selected conditional coefficients on a common distance-equivalent scale.⁴ Because distance enters negatively in all four indices, a negative entry means that the covariate change raises the fitted index by as much as a shorter distance would, while a positive entry means that it lowers the index by as much as a longer distance would. A one-standard-deviation increase in working hours—8.68 hours in the sample—is equivalent to increasing distance by 3.7 percent for sorting and 8.1 percent for doctor acceptance, but to decreasing distance by 58.2 percent for screening and 51.3 percent for post acceptance. The doctor-side signs are consistent with longer shifts being less attractive to doctors; the opposite screening and post-side signs are consistent with agents directing more effort toward, and facilities accepting more readily for, longer shifts.

Most match and prior-relationship variables move in the expected direction. Turning preference match from zero to one is equivalent to reducing distance by 23.6, 15.6, 49.6, and 40.1 percent in sorting, screening, doctor acceptance, and post acceptance, respectively. One stand-

⁴For each index g , let $\hat{\beta}_{xg}$ denote the estimated coefficient on covariate x , let $\hat{\beta}_{dg}$ denote the estimated coefficient on standardized log distance, and let $\sigma_{\log d} = 0.899556$ denote the sample standard deviation of unstandardized log distance. Continuous covariates are standardized before estimation, so $\hat{\beta}_{xg}$ is the fitted-index change from a one-standard-deviation increase; indicators remain unstandardized, so it is the change from zero to one. I choose the equivalent distance change by solving $\hat{\beta}_{xg} = \hat{\beta}_{dg} \Delta \log d / \sigma_{\log d}$. The table therefore reports $100\{\exp(\sigma_{\log d} \hat{\beta}_{xg} / \hat{\beta}_{dg}) - 1\}$, rounded to one decimal. For example, the hours and log-distance coefficients in the sorting index are -0.04076 and -1.02076 , so the transformation gives $100\{\exp[0.899556(-0.04076 / -1.02076)] - 1\} = 3.66$, reported as 3.7.

Table 4. Distance-Equivalent Changes for Selected Covariates

	Sorting	Screening	Doctor acc.	Post acc.
Hours	3.7	-58.2	8.1	-51.3
Posted salary	-22.4	-11.0	-12.3	-7.0
Preference match	-23.6	-15.6	-49.6	-40.1
Day match	-42.3	4.2	-59.8	-33.3
Specialty match	-38.4	-17.1	-45.0	-35.2
Prior approaches (past 365 days)	-16.9	-8.0	-11.4	-12.0
Prior contract	-6.8	-5.5	-0.9	-1.5

Notes: Entries report the percentage change in distance producing the same fitted-index change as a one-sample-SD continuous change or a zero-to-one indicator change. Negative entries move the index like shorter distance; positive entries move it like longer distance. These are point transformations only; no uncertainty measures or significance symbols are supplied.

ard deviation more prior approaches—1.92 approaches in the sample—acts like an 8.0–16.9 percent reduction in distance across the four margins. We find some conditional signs remain counterintuitive: a day match acts like 4.2 percent greater distance in screening, even though it acts like shorter distance in the other three margins.

5 Policy Exercises

The policy exercises change how the platform recommends jobs to doctors. They are motivated by the platform’s consideration of an algorithmic recommendation system and ask three broad questions. First, should algorithmic recommendations operate alongside doctors’ own search and applications, or should they replace doctor search entirely? Second, if doctor search remains available, what should the algorithm prioritize: spreading recommendation opportunities more evenly across posts, generating greater immediate gains from matches, or creating greater long-run value after doctors and posts adjust? Third, should the platform increase the number of recommendations rather than only change who receives them?

Section 5.1 first explains how counterfactual recommendation policies are constructed. The next three subsections answer the three questions in the same order. Section 5.2 compares keeping doctor search with replacing it and shows why the model favors retaining decentralized search. Section 5.3 then conditions on doctor search remaining available, compares what the algorithm should prioritize, and shows how the long-run rule changes who receives recommendations. Finally, Section 5.4 keeps doctor search available and asks what the platform gains from making more recommendations, separates that change from reallocating recommendations across doctors and posts, and compares the gains from redesign with what could be achieved by expanding the market itself.

5.1 Policy Classes and Objectives

All counterfactual policies choose the same object. Let

$$A = (A_{ij})_{i \in I, j \in J}, \quad A_{ij} \geq 0,$$

where A_{ij} is the expected number of monthly contacts that the platform assigns to doctor–post pair (i, j) through recommendations. Equivalently, A_{ij}/K_i is the recommendation rate for one slot of doctor i . A policy class determines which matrices A are feasible and whether doctor search remains available. A policy objective determines how the platform chooses A within that class. I introduce these two ingredients in turn and then describe how the resulting policies are computed and compared.

5.1.1 Policy Class

For each policy class, the allocation exercises keep the expected number of contacts that the platform assigns to each doctor at the level estimated for the current market and change only which posts receive those contacts. This is because a policy that sharply changes this number for an individual doctor would be difficult to justify in practice. Instead, I do not impose the same requirement on posts and the number of platform-assigned contacts received by an individual post may therefore change substantially, subject to the pair-specific limits below.

To implement these comparisons, I use fitted current contacts to define pair-specific upper bounds on the policy matrix. Let

$$\hat{S}_{ij} = K_i \lambda_{ij}^{\text{self}}(\hat{\alpha}_i^D) \quad \text{and} \quad \hat{A}_{ij} = K_i \lambda_{ij}^{\text{agency}}(\hat{\alpha}_j^P)$$

denote the fitted monthly numbers of self-initiated and agency-initiated contacts under the current policy. For the one-month horizon, define the pair-specific reference level

$$\overline{M}_{ij} = \max\{K_i, \hat{S}_{ij} + \hat{A}_{ij}\}.$$

This reference allows one contact per doctor slot for pair (i, j) unless the fitted current total is larger, so it never excludes the fitted current allocation. I define the portion that remains available for platform recommendations after accounting for fitted self-initiated contacts as

$$\overline{A}_{ij} = \max\{\overline{M}_{ij} - \hat{S}_{ij}, 0\}.$$

The two policy classes use these quantities differently: $\overline{A}_{ij}$ caps recommendations when doctor search remains available, whereas $\overline{M}_{ij}$ caps recommendations when all contacts are assigned by the platform.

Search-retained class. The *search-retained* class changes platform recommendations while leaving doctor-initiated search governed by the estimated behavioral rule. Let b_i denote the expected monthly number of contacts that the platform assigns to doctor i through recommendations. For a feasible vector $b = (b_i)_{i \in I}$, the class has feasible set

$$\mathcal{B}^{\text{retain}}(b) = \left\{ A \geq 0 : \sum_j A_{ij} = b_i \ \forall i, \quad A_{ij} \leq \bar{A}_{ij} \right\}. \quad (8)$$

The baseline exercise uses each doctor's fitted current recommendation budget, $b_i^{\text{agency}} = \sum_j \hat{A}_{ij}$. For a chosen A , the platform replaces the behavioral agency-intensity equation with the forced rate A_{ij}/K_i . Doctor search continues to satisfy

$$M_{ij}^{\text{self}}(\alpha; K) = K_i \lambda_{ij}^{\text{self}}(\alpha_i^D).$$

Continuation values and acceptance therefore solve

$$\begin{aligned} (r + \delta_i^a) \alpha_i^D &= \sum_j \left[\lambda_{ij}^{\text{self}}(\alpha_i^D) \psi_{ij, \text{self}}^D(\alpha) + \frac{A_{ij}}{K_i} \psi_{ij, \text{agency}}^D(\alpha) \right], \\ (r + w_j) \alpha_j^P &= \sum_i \left[K_i \lambda_{ij}^{\text{self}}(\alpha_i^D) \psi_{ij, \text{self}}^P(\alpha) + A_{ij} \psi_{ij, \text{agency}}^P(\alpha) \right]. \end{aligned}$$

Write $\alpha^{\text{retain}}(A; K)$ for the solution. Thus A is the only object assigned directly by the platform; doctor search, acceptance, and continuation values respond in equilibrium.

Full-replacement class. The *full-replacement* class removes doctor-initiated search and assigns all contacts through platform recommendations. Its feasible set is

$$\mathcal{B}^{\text{replace}}(b) = \left\{ A \geq 0 : \sum_j A_{ij} = b_i \ \forall i, \quad A_{ij} \leq \bar{M}_{ij} \right\}.$$

The baseline exercise uses each doctor's fitted current total-contact budget, $b_i^{\text{total}} = \sum_j (\hat{S}_{ij} + \hat{A}_{ij})$. A candidate matrix A sets $M^{\text{self}} = 0$ and $M^{\text{agency}} = A$, so continuation values solve

$$\begin{aligned} (r + \delta_i^a) \alpha_i^D &= \sum_j \frac{A_{ij}}{K_i} \psi_{ij, \text{agency}}^D(\alpha), \\ (r + w_j) \alpha_j^P &= \sum_i A_{ij} \psi_{ij, \text{agency}}^P(\alpha). \end{aligned}$$

Write the solution as $\alpha^{\text{replace}}(A; K)$.

5.1.2 Policy Objective

For a fixed policy class $k \in \{\text{retain}, \text{replace}\}$ and feasible b , I compare three ways of choosing $A \in \mathcal{B}^k(b)$. All three hold fixed the policy class and the number of recommendations assigned to each doctor. They differ only in how those recommendations are distributed across posts.

Equality: even allocation. The first rule spreads each doctor’s recommendations as evenly as the pair-specific limits allow and is reported as *Random*.

Let $\bar{C}^{\text{retain}} = \bar{A}$ and $\bar{C}^{\text{replace}} = \bar{M}$. The Random allocation is

$$R_{ij}^k(b) = \min\{\bar{C}_{ij}^k, \tau_i^k(b)\}, \quad \sum_j R_{ij}^k(b) = b_i,$$

where $\tau_i^k(b)$ is chosen so that doctor i receives exactly b_i recommendations. It is a benchmark for an even allocation, not the solution to a separate fairness-maximization problem. For the baseline comparison, write $R^{\text{retain}} = R^{\text{retain}}(b^{\text{agency}})$ and $R^{\text{replace}} = R^{\text{replace}}(b^{\text{total}})$.

Short run: immediate match gains. The short-run problem allocates recommendations to maximize immediate expected bilateral surplus at the fitted continuation values. To state this problem, let the route-specific contact matrices realized at continuation vector α be

$$M_{ij}^{c,k}(A; \alpha) = \begin{cases} K_i \lambda_{ij}^{\text{self}}(\alpha_i^D), & k = \text{retain}, c = \text{self}, \\ A_{ij}, & k = \text{retain}, c = \text{agency}, \\ 0, & k = \text{replace}, c = \text{self}, \\ A_{ij}, & k = \text{replace}, c = \text{agency}. \end{cases}$$

The objective is

$$S^k(A) = \sum_{i,j,c} M_{ij}^{c,k}(A; \hat{\alpha}) \left\{ \psi_{ijc}^D(\hat{\alpha}) + \psi_{ijc}^P(\hat{\alpha}) \right\}.$$

The short-run rule maximizes $S^k(A)$ over $A \in \mathcal{B}^k(b)$. It holds opportunity costs and acceptance fixed at $\hat{\alpha}$. In the search-retained class it also holds doctor contacts at $\hat{S}$, so only the recommendation matrix changes immediately.

Long run: value after equilibrium adjustment. The long-run rule accounts for how a recommendation allocation changes the equilibrium. For each candidate A , I re-solve acceptance and continuation values. In the search-retained class, this calculation also updates the self-search contact rates $\lambda^{\text{self}}(\alpha^D)$. I then add the resulting continuation values across all doctor slots and posts. I refer to this sum as total user value in the model:

$$W^k(A; K) = \sum_i K_i \alpha_i^{D,k}(A; K) + \sum_j \alpha_j^{P,k}(A; K). \quad (9)$$

The first term counts one doctor-side continuation value for each of doctor i ’s K_i slots, and the second counts one post-side continuation value for each post. The long-run problem maximizes $W^k(A; K)$ over $A \in \mathcal{B}^k(b)$. I report the best allocation found by the finite search described in the footnote.⁵

⁵For each policy class, I evaluate a finite set of starting allocations and begin from the one with the highest value of W^k . I then run at most eight ascent iterations. At each iteration, a finite line search accepts an allocation

Comparison across the three rules. The three rules prioritize different aspects of the allocation. Random spreads recommendations as evenly as the pair-specific limits permit. The short-run rule maximizes immediate expected bilateral surplus at the fitted continuation values. The long-run rule instead seeks the highest total user value after doctors and posts respond in equilibrium, as defined in equation (9). I compare the rules along two dimensions: how evenly they distribute post-side outcomes and how much value they generate according to the two objective functions. Because posts are the institutions that pay for the platform’s service, the platform must also avoid recommendation policies that distribute the benefits of that service too unevenly across them; I use the post-side access index to capture this practical equity concern.

To compare the distribution of outcomes across posts, I define a post-specific outcome for each objective. In the short-run comparison, $y_j^{SR,k}(A)$ is post j ’s immediate expected surplus flow at the fitted continuation values. In the long-run comparison, $y_j^{LR,k}(A)$ is post j ’s equilibrium continuation value:

$$y_j^{SR,k}(A) = \sum_{i,c} M_{ij}^{c,k}(A; \hat{\alpha}) \psi_{ijc}^P(\hat{\alpha}), \quad y_j^{LR,k}(A) = \alpha_j^{P,k}(A; K).$$

For each post, I compare its outcome under A with its outcome under the corresponding Random allocation. For $q \in \{SR, LR\}$, the resulting post-side access index is

$$\mathcal{A}^{q,k}(A) = 100 \left[1 - \text{Gini}_{j \in J} \left\{ \frac{y_j^{q,k}(A)}{y_j^{q,k}(R^k)} \right\} \right],$$

provided $y_j^{q,k}(R^k) > 0$. Random scores 100 because, when $A = R^k$, every ratio inside the Gini coefficient equals one and the Gini coefficient of this constant vector is zero. A lower score means that posts’ outcomes change by more unequal proportions relative to Random. The index is neither the share of posts receiving recommendations nor a measure of inequality in the raw levels of post outcomes.

I also evaluate each displayed allocation under both objective functions. Within each policy class, the short-run objective is reported as a percentage of the value attained by the short-run maximizing rule, so that rule equals 100. The long-run objective is reported as a percentage of the value attained by the best-found long-run rule, so that rule also equals 100.

Within each policy class, I therefore report Random, the short-run maximizing rule, and the best-found long-run rule, together with the current hybrid as a comparison. In the full-replacement panel, the current hybrid is an external two-route comparator rather than an element of $\mathcal{B}^{\text{replace}}(b)$. I evaluate its fitted outcomes using the same full-replacement benchmarks.

only if it raises W^k . The search stops at an allocation fixed point, after two consecutive iterations without a relative gain above 10^{-4} , or after the eighth iteration. Every accepted allocation must satisfy the equilibrium equations with a maximum residual no greater than 10^{-5} . Because this finite search does not provide a global optimality certificate, I refer to its output as the “best-found” long-run design. All reported values are conditional on the maintained equilibrium branch.

Its normalized objective values can therefore exceed 100: in that panel, 100 is the value attained within the full-replacement policy class, not an upper bound for the external current hybrid.

5.2 Should Doctor Search Remain Available?

This subsection answers the first question: should doctor search remain available alongside algorithmic recommendations? The answer is yes. When doctor search remains available, doctors adjust the number of contacts they initiate as continuation values change. Doctors can therefore respond to the platform’s recommendation policy, and long-run performance remains high across different recommendation rules. Once doctor search is removed, the platform’s choice of objective becomes much more consequential. Because the platform must balance post-side equity, immediate match gains, and long-run value rather than pursue a single objective, this ability to perform well across recommendation rules makes retaining doctor search especially valuable.

Table 5 shows this contrast. With doctor search retained, Random, the short-run rule, and the long-run rule attain 97.97, 99.21, and 100 percent of the best-found long-run value, respectively. The platform assigns the same 3,051.4 contacts through recommendations under all three rules, while equilibrium doctor-initiated contacts range from 3,722.4 to 3,886.8. Under full replacement, by contrast, the short-run rule attains only 53.27 percent of the best-found long-run value and a long-run access score of 21.66, compared with 80.53 under the long-run rule. Removing doctor search therefore makes both value and post-side access much more sensitive to what the algorithm targets. Appendix Figure C.2 plots the corresponding access–value comparison. Once the platform assigns all contacts, the short-run and long-run rules separate sharply in both post-side access and the value they generate.

5.3 What Should the Algorithm Prioritize When Doctor Search Remains?

Given that doctor search should remain available, this subsection answers the second question: what should the recommendation algorithm prioritize? Because the platform pursues several objectives at once, the relevant comparison is not simply whether each rule performs well on the criterion it was designed to maximize. The more important question is how well it performs on the other criteria that the platform also values. By this standard, the best-found long-run rule provides the most robust design.

Panel A of Table 5 shows this result. The long-run rule attains 100 percent of the best-found long-run value while preserving 98.34 percent of the maximum short-run objective. The short-run rule, by comparison, attains 100 percent of the short-run objective and 99.21 percent of the best-found long-run value. Moving from the short-run rule to the long-run rule therefore sacrifices only 1.66 percentage points of immediate match gains. At the same time, it raises the

Table 5. Access, Value, and Realized Exposure under Alternative Policies

Policy rule	Agency	Realized self	Realized total	Short access	Long access	Short objective	Long value
<i>Panel A: Doctor search retained; agency screening redesigned</i>							
Random	3,051.4	3,752.4	6,803.9	100.00	100.00	83.16	97.97
Current hybrid	3,051.4	3,813.5	6,864.9	59.94	88.78	84.52	97.93
Short-run best found	3,051.4	3,722.4	6,773.8	52.28	80.93	100.00	99.21
Long-run best found	3,051.4	3,886.8	6,938.2	60.34	81.27	98.34	100.00
<i>Panel B: Doctor search replaced; all contacts assigned by platform</i>							
Random	6,864.9	0.0	6,864.9	100.00	100.00	41.78	90.55
Current hybrid	3,051.4	3,813.5	6,864.9	46.30	75.91	114.86	110.32
Short-run best found	6,864.9	0.0	6,864.9	10.81	21.66	100.00	53.27
Long-run best found	6,864.9	0.0	6,864.9	53.57	80.53	71.39	100.00

Notes: Agency is the platform-controlled recommendation count. Realized self is the equilibrium doctor-initiated count, and realized total is their sum. The displayed Realized self and Realized total columns are the long-run equilibrium counts for each rule. In Panel A, doctor search is recomputed from continuation values under every screening matrix for the long-run evaluation; it is not held fixed. The short-run objective instead holds baseline self-search contacts and baseline continuation and surplus terms fixed. In Panel B, realized self contacts are zero for full-replacement candidates; the current hybrid is a comparison benchmark outside that candidate class. Access is the random-normalized post-side index. The short objective and long value are normalized within policy class. Best-found is under maintained equilibrium selection and does not claim a global optimum. Agency budget and agency decision-pair caps are feasibility gates. Any realized-total overage created by endogenous doctor search is reported as a non-gating diagnostic.

short-run access score from 52.28 to 60.34 and the long-run access score from 80.93 to 81.27. Accounting for equilibrium responses thus improves long-run value and produces more even post-side outcomes than pure short-run targeting, at little cost in immediate gains. Appendix Figure C.1 plots the same comparison with doctor search retained: the long-run rule stays close to the highest value in both panels and gives posts more even outcomes than the short-run rule.

Random remains the relevant alternative when post-side equity receives much greater weight. It raises both access scores to 100 and still preserves 97.97 percent of the best-found long-run value, but it attains only 83.16 percent of the maximum short-run objective.⁶ No rule therefore dominates when post-side equity is treated as a separate objective. Nevertheless, we consider the long-run rule is the natural default for a multi-objective platform: it nearly maximizes immediate match gains, attains the highest best-found long-run value, and preserves more

⁶Random remains strong in the long-run metric for two reasons. First, valuable future matching opportunities are not concentrated in a small set of posts, so spreading recommendations broadly preserves much of aggregate continuation value. Second, retained doctor search provides an endogenous adjustment margin: doctors redirect their own contacts as continuation values change, partially compensating for a diffuse recommendation allocation. This does not mean that all doctor-post pairs are equally valuable, as shown by Random attaining only 83.16 percent of the maximum short-run objective. Nor does it imply that doctors are homogeneous, because each doctor's total number of recommendations is held fixed in this comparison.

Table 6. How the Best-Found Long-Run Rule Retargets Recommendations

Recommendation-weighted attribute	Current	Best found	Change
Job duration (hours)	12.82	5.23	−7.60
Mean log distance	3.193	2.854	−0.339
Preference match (%)	14.65	19.59	+4.94 pp
Day match (%)	8.61	13.87	+5.26 pp
Specialty match (%)	7.38	10.62	+3.24 pp
Distance-projection coefficient	−2.558	−10.570	−8.012

Notes: Entries weight observed doctor–post attributes by platform-controlled recommendations. Current refers to fitted agency recommendations, and Best found refers to the best-found long-run recommendation matrix when doctor search remains available. Each doctor’s total number of recommendations is held at its fitted current level. Equilibrium doctor search is recomputed when evaluating the policy but is not used as a weight in this table. The distance row is a projection of within-doctor recommendation shares on log distance with doctor and post fixed effects.

post-side access than the short-run rule.

Table 6 turns the long-run rule into operational guidance by comparing current and best-found recommendations with doctor search retained. Each doctor’s recommendation total is fixed, so the differences show which posts receive the existing recommendations.

The best-found long-run rule shifts recommendations toward shorter, closer, and more compatible jobs. The recommendation-weighted duration of jobs falls from 12.82 to 5.23 hours. Mean log distance falls from 3.193 to 2.854, which corresponds to a decline in geometric-mean distance from 24.4 to 17.4 kilometers. The shares of recommendations involving a preference match, a day match, and a specialty match rise by 4.94, 5.26, and 3.24 percentage points, respectively. The stronger tilt toward nearby posts remains after accounting for stable differences across doctors and posts: the distance-projection coefficient becomes more negative, moving from −2.558 to −10.570. For implementation, these results point toward a clear revision of the platform’s ranking rule: place greater weight on proximity, shorter jobs, and observed preference, schedule, and specialty compatibility rather than simply reproducing current agent screening.⁷

This exercise changes which posts receive the existing recommendations. The remaining question is whether the platform can create larger gains by increasing the number of recommendations rather than only changing their allocation.

⁷These statistics compare two model-computed recommendation matrices. They are not causal effects of changing a post’s hours, location, or compatibility, and they do not imply that a score using only these attributes would reproduce the best-found allocation. The comparison also does not propagate parameter-estimation or policy-search uncertainty.

5.4 Should the Platform Make More Recommendations?

I organize this exercise around three sequential policy margins. First, the platform increases the total number of recommendations. Second, holding that larger total fixed, it assigns the same expected number of recommendations to every doctor. Third, holding both the aggregate volume and each doctor’s recommendation total fixed, it redesigns which posts receive those recommendations. These margins correspond to three distinct operational decisions: how many recommendations the platform makes, whether doctors receive equal recommendation opportunities, and how the algorithm allocates recommendations across doctor–post pairs. The decomposition therefore separates the gains from recommendation expansion, doctor-side equalization, and pairwise redesign.

All three margins raise aggregate continuation value under the long-run objective. With doctor search retained, doubling the fitted current recommendation profile raises aggregate continuation value by 2.23 percent after equilibrium adjustment. Holding the doubled aggregate volume fixed and equalizing recommendation totals across doctors produces an additional gain of 10.25 percent. Finally, holding those equal totals fixed and redesigning which posts receive the recommendations produces a further gain of 9.58 percent. Thus, in this exercise, the largest gains come not from increasing recommendation volume alone, but from changing how recommendations are distributed across doctors and doctor–post pairs.

To construct the first step, I start from doctor i ’s fitted current recommendation total,

$$b_i^{\text{agency}} = \sum_j \hat{A}_{ij},$$

in the feasible set in equation (8). The doubled profile assigns doctor i a recommendation total of $2b_i^{\text{agency}}$. For the second step, I keep the same doubled aggregate volume but assign every doctor the same expected number of recommendations:

$$b_i^{\text{eq}} = \frac{2 \sum_{\ell} b_{\ell}^{\text{agency}}}{|I|}.$$

The first two profiles proportionally rescale each doctor’s fitted current allocation across posts, using water filling when a pair-specific upper bound binds. The equalization step applies only to recommendations assigned by the platform; endogenous doctor search can still make total contacts differ across doctors. For the third step, I hold b^{eq} fixed and search over

$$A \in \mathcal{B}^{\text{retain}}(b^{\text{eq}})$$

for the best-found long-run allocation of each doctor’s recommendations across posts.

Table 7 reports the value gains from these three sequential changes. Doubling the current recommendation profile raises aggregate continuation value from 16,252.23 to 16,614.68, a gain of 362.45, or 2.23 percent. Equalizing recommendation totals across doctors then raises it to

Table 7. Recommendation Expansion, Equalization across Doctors, and Pairwise Redesign

Policy step	Platform recommendations	Doctor search	Total contacts	Aggregate continuation value	Change (%)
Current hybrid	3,051.4	3,813.5	6,864.9	16,252.2	—
Double current recommendation profile	6,102.8	3,902.6	10,005.5	16,614.7	+2.23
Equalize recommendation totals across doctors	6,102.8	4,225.5	10,328.4	18,318.4	+10.25
Best-found pair allocation	6,102.8	4,415.2	10,518.0	20,072.5	+9.58

Notes: Aggregate continuation value is model-implied; each percentage is relative to the preceding row. Platform recommendations are fixed by the policy, while doctor-initiated contacts respond endogenously through the re-solved continuation values. Total contacts therefore need not equal the platform’s recommendation target. Recommendation totals and pair-specific upper bounds determine feasibility.

18,318.39, an additional gain of 1,703.71, or 10.25 percent. Holding those equal totals fixed and redesigning which posts receive the recommendations raises it to 20,072.48, a further gain of 1,754.09, or 9.58 percent. Appendix Figure C.3 displays the same sequential decomposition.

The 1,754.09 gain from pairwise redesign is difficult to interpret in isolation. From the platform’s perspective, improving the recommendation allocation is an alternative to expanding the market by attracting additional doctors or posts. I therefore ask how much targeted market expansion would be required to generate the same increase in aggregate continuation value. This comparison places algorithmic redesign and market expansion on the same model-implied value scale.

For this benchmark, doctor-side expansion increases the number of slots available from selected doctor types. On the post side, let L_j denote the number of identical copies of observed post j , with $L_j = 1$ in the observed market. Aggregate contacts directed to post type j are divided across its copies, so each copy receives route- c contact flow M_{ij}^c/L_j . The corresponding long-run value is

$$W^{\text{retain}}(A; K, L) = \sum_i K_i \alpha_i^{D, \text{retain}}(A; K, L) + \sum_j L_j \alpha_j^{P, \text{retain}}(A; K, L).$$

I rank doctors by $\hat{\alpha}_i^D$ and observed posts by $\hat{\alpha}_j^P$, and add stock to the top decile on each side. Each addition is distributed across the selected types in proportion to their baseline stocks. During this exercise, I hold the platform recommendation matrix fixed and recompute doctor search as market stocks and continuation values change. I solve the equilibrium over a seven-by-seven grid, measuring each addition as a percentage of the corresponding aggregate baseline stock.

With post stock fixed, reproducing the redesign gain requires additional capacity from top-decile doctor types equal to 9.31 percent of aggregate baseline doctor-slot stock. With doctor stock fixed, it instead requires top-decile post stock equal to 26.80 percent of aggregate baseline post stock. If top-decile post stock first increases by 20 percent, the additional doctor-slot capacity required to reproduce the gain falls to 1.42 percent. Table 8 summarizes these comparisons, while Appendix Figure C.5 shows the combinations of doctor and post additions across

Table 8. Market-Composition Equivalents of the Pairwise-Redesign Gain with Doctor Search Retained

Policy class	Redesign increment	Doctor-slot stock added	Post stock added	Doctor slots with +20% post stock	Closed by +40% post stock
Doctor search retained	1,754.1	9.31%	26.80%	1.42%	127.97%

Notes: The redesign increment is the best-found gain after recommendation volume is doubled and recommendation totals are equalized across doctors. Market composition adds aggregate baseline stock to top-decile types and re-solves equilibrium, with doctor search endogenous throughout. A one-sided addition reported as above 40 percent does not reach the target on the evaluated grid.

the evaluated grid.

These results show that pairwise redesign is economically important relative to market expansion. Reallocating a fixed number of recommendations across posts generates as much long-run value as a sizable, targeted expansion of either side of the market. They also reveal complementarity between the two sides: expanding post capacity sharply reduces the additional doctor capacity needed to attain the same value gain. Thus, improving the recommendation algorithm can partly substitute for one-sided market expansion, while expanding both sides together increases the value of additional capacity.

References

- Adachi, Hiroyuki.** 2003. “A Search Model of Two-Sided Matching under Nontransferable Utility.” *Journal of Economic Theory*, 113(2): 182–198.
- Agarwal, Nikhil.** 2015. “An Empirical Model of the Medical Match.” *American Economic Review*, 105(7): 1939–1978.
- Belot, Michèle, Philipp Kircher, and Paul Muller.** 2019. “Providing Advice to Job-seekers at Low Cost: An Experimental Study on Online Advice.” *Review of Economic Studies*, 86(4): 1411–1447.
- Biega, Asia J., Krishna P. Gummadi, and Gerhard Weikum.** 2018. “Equity of Attention: Amortizing Individual Fairness in Rankings.” 405–414.
- Burdett, Ken, and Melvyn G. Coles.** 1997. “Marriage and Class.” *Quarterly Journal of Economics*, 112(1): 141–168.
- Chade, Hector, Jan Eeckhout, and Lones Smith.** 2017. “Sorting through Search and Matching Models in Economics.” *Journal of Economic Literature*, 55(2): 493–544.

- Chen, Kuan-Ming, Yu-Wei Hsieh, and Ming-Jen Lin.** 2023. “Reducing Recommendation Inequality via Two-Sided Matching: A Field Experiment of Online Dating.” *International Economic Review*, 64(3): 1201–1221.
- Chen, Nan, and Hsin-Tien Tsai.** 2024. “Steering via Algorithmic Recommendations.” *RAND Journal of Economics*, 55(4): 501–518.
- Eeckhout, Jan, and Philipp Kircher.** 2010. “Sorting versus Screening: Search Frictions and Competing Mechanisms.” *Journal of Economic Theory*, 145(4): 1354–1385.
- Fradkin, Andrey.** 2017. “Search, Matching, and the Role of Digital Marketplace Design in Enabling Trade: Evidence from Airbnb.” SSRN Working Paper.
- Galichon, Alfred, Scott Duke Kominers, and Simon Weber.** 2019. “Costly Concessions: An Empirical Framework for Matching with Imperfectly Transferable Utility.” *Journal of Political Economy*, 127(6): 2875–2925.
- Gong, Jing.** 2016. “A Structural Two-Sided Matching Model of Online Labor Markets.” SSRN Working Paper.
- Hitsch, Gunter J., Ali Hortaçsu, and Dan Ariely.** 2010. “Matching and Sorting in Online Dating.” *American Economic Review*, 100(1): 130–163.
- Horton, John J.** 2017. “The Effects of Algorithmic Labor Market Recommendations: Evidence from a Field Experiment.” *Journal of Labor Economics*, 35(2): 345–385.
- Immorlica, Nicole, Brendan Lucier, Vahideh Manshadi, and Alexander Wei.** 2023. “Designing Approximately Optimal Search on Matching Platforms.” *Management Science*, 69(8): 4609–4626.
- Lauermann, Stephan, and Georg Nöldeke.** 2025. “Matching with Frictions.” In *Handbook of the Economics of Matching*. Vol. 2, Chapter 7, 579–642. Elsevier B.V.
- Le Barbanchon, Thomas, Lena Hensvik, and Roland Rathelot.** 2023. “How Can AI Improve Search and Matching? Evidence from 59 Million Personalized Job Recommendations.” SSRN Working Paper.
- Manshadi, Vahideh, Scott Rodilitz, Daniela Saban, and Akshaya Suresh.** 2025. “Online Algorithms for Matching Platforms with Multichannel Traffic.” *Management Science*, 71(9): 7674–7691.
- Moen, Espen R.** 1997. “Competitive Search Equilibrium.” *Journal of Political Economy*, 105(2): 385–411.

- Shimer, Robert, and Lones Smith.** 2000. “Assortative Matching and Search.” *Econometrica*, 68(2): 343–369.
- Shi, Peng.** 2023. “Optimal Matchmaking Strategy in Two-Sided Marketplaces.” *Management Science*, 69(3): 1323–1340.
- Shi, Peng.** 2025. “Optimal Match Recommendations in Two-Sided Marketplaces with Endogenous Prices.” *Management Science*, 71(9): 7431–7448.
- Sürer, Özge, Robin Burke, and Edward C. Malthouse.** 2018. “Multistakeholder Recommendation with Provider Constraints.” 54–62.
- Todorov, Emanuel.** 2006. “Linearly Solvable Markov Decision Problems.” Vol. 19. MIT Press.
- Wang, Yichen, Grady Williams, Evangelos Theodorou, and Le Song.** 2017. “Variational Policy for Guiding Point Processes.” Vol. 70 of *Proceedings of Machine Learning Research*, 3684–3693. PMLR.

A Model Details and Proofs

A.1 Route-Specific Closed Forms

For route c , write $\mu_{ijc}^D := u_{ijc}^D$ and $\mu_{jic}^P := u_{jic}^P$ for the systematic payoffs parameterized in Section 4.1. If ε is logistic with scale ζ and location zero, then

$$\mathbb{E}[(\mu + \varepsilon - \alpha)_+] = \zeta \log(1 + \exp\{(\mu - \alpha)/\zeta\}).$$

Consequently,

$$\begin{aligned} q_{ijc}^D(\alpha) &= \zeta^D \log\left(1 + \exp\{(\mu_{ijc}^D - \alpha_i^D)/\zeta^D\}\right), \\ q_{ijc}^P(\alpha) &= \zeta^P \log\left(1 + \exp\{(\mu_{jic}^P - \alpha_j^P)/\zeta^P\}\right), \\ \psi_{ijc}^D(\alpha) &= \frac{q_{ijc}^D(\alpha)}{1 + \exp\{(\alpha_j^P - \mu_{jic}^P)/\zeta^P\}}, \quad \psi_{ijc}^P(\alpha) = \frac{q_{ijc}^P(\alpha)}{1 + \exp\{(\alpha_i^D - \mu_{ijc}^D)/\zeta^D\}}. \end{aligned} \tag{10}$$

These expressions make clear why route rates cannot be collapsed once either initiator-specific acceptance intercept difference is nonzero. For example,

$$\lambda_{ij}^{\text{self}} \psi_{ij,\text{self}}^D + \lambda_{ij}^{\text{agency}} \psi_{ij,\text{agency}}^D \neq (\lambda_{ij}^{\text{self}} + \lambda_{ij}^{\text{agency}}) \psi_{ij}^D$$

for any route-common ψ_{ij}^D in general. Setting both χ terms to zero recovers the nested route-common acceptance specification.

A.2 Auxiliary Representation of Behavioral Intensities

The behavioral exposure equations admit the relative-entropy representation in equation (1). For a Poisson intensity λ and baseline $\bar{\lambda} > 0$, the KL divergence rate defined in the main text is

$$D_{\text{KL}}(\lambda \parallel \bar{\lambda}) = \lambda \log(\lambda/\bar{\lambda}) - \lambda + \bar{\lambda}.$$

Its derivative with respect to λ is $\log(\lambda/\bar{\lambda})$. The auxiliary objective

$$\lambda(s - \omega) - D_{\text{KL}}(\lambda \parallel \bar{\lambda})$$

is strictly concave for $\lambda > 0$, and its unique first-order solution is

$$\log \lambda = \log \bar{\lambda} + s - \omega.$$

These expressions use the maintained normalization $\zeta^{\text{self}} = \zeta^{\text{agency}} = 1$ for the two exposure routes. Hence the relevant continuation value enters each behavioral log-intensity with coefficient minus one, not with an estimated route-specific loading. For the doctor-search route, $\omega = \alpha_i^D$; for behavioral agent screening at the estimated current-market equilibrium, $\omega = \alpha_j^P$. A constant $\log \bar{\lambda}$ is observationally part of the route's exposure intercept, so the implementation does not estimate a separate baseline-rate parameter.

This representation supplies intuition for the log-intensity specification, not a second payoff term in the stationary-renewal problem. In particular, the KL expression is absent from the continuation-value equations and from the counterfactual value objective. In the search-retained counterfactual the platform also replaces the behavioral agency first-order condition with a fixed agency contact matrix, while the doctor-search first-order condition continues to determine $\lambda^{\text{self}}(\alpha^D)$.

A.3 Proof of Proposition 1

Proof. Characterisation. Let (a, α) be a fixed-rate stationary-renewal equilibrium. Because the payoff shocks are continuous, ties occur with probability zero, and optimality requires the threshold rules in Section 3.1. Consider one waiting slot of doctor i over an interval of length dt . A route- c exposure to post j occurs with probability $\lambda_{ij}^c dt + o(dt)$. Relative to continued waiting, its expected value is

$$\mathbb{E}[a_{jic}^P(U_{ij}^c - \alpha_i^D)_+] = \psi_{ijc}^D(\alpha).$$

Discounting and abandonment reduce the waiting value at rate $r + \delta_i^a$. Stationary renewal leaves the environment after every event unchanged. Thus

$$\alpha_i^D = (1 - (r + \delta_i^a)dt)\alpha_i^D + \sum_{j,c} \lambda_{ij}^c dt \psi_{ijc}^D(\alpha) + o(dt),$$

which gives the doctor equation in (5). The post equation follows in the same way, except that the K_i independent slots generate a total route- c flow $K_i \lambda_{ij}^c$ from doctor i to post j . Hence every equilibrium continuation vector is a fixed point. Conversely, a fixed point together with the associated threshold rules satisfies optimal acceptance and value consistency, and therefore is a stationary-renewal equilibrium.

Existence. For fixed nonnegative finite route rates, every function ψ_{ijc}^D and ψ_{ijc}^P in (10) is continuous and nonnegative. It is also nonincreasing in every continuation-value argument on which it depends. Therefore $\Phi_K(\cdot; \lambda)$ is continuous and, for all $\alpha \geq 0$,

$$0 \leq \Phi_K(\alpha; \lambda) \leq \Phi_K(0; \lambda)$$

componentwise. Define the box

$$B = \prod_k [0, \Phi_{K,k}(0; \lambda)].$$

It is nonempty, compact, and convex, and $\Phi_K(B; \lambda) \subseteq B$. Brouwer's fixed-point theorem gives at least one fixed point in B . $\square$

Under the behavioral rule (2), route rates depend on the relevant continuation value. Substituting $\Gamma(\alpha; s)$ into Φ_K preserves continuity and nonnegativity. Both Γ and the expected-gain terms are componentwise nonincreasing on the nonnegative orthant, so

$$0 \leq \Phi_K(\alpha; \Gamma(\alpha; s)) \leq \Phi_K(0; \Gamma(0; s))$$

componentwise. The box whose upper corner is the last expression is invariant, so Brouwer's theorem establishes existence for the behavioral system (6). The search-retained counterfactual is a hybrid instance of the same argument: the agency matrix is fixed, while $\lambda^{\text{self}}(\alpha^D)$ remains continuous, nonnegative, and bounded above by its value at zero. Only the full-replacement benchmark holds both route matrices fixed while continuation values and acceptance are resolved.

A.4 A Sufficient Condition for Fixed-Exposure Uniqueness

Define the zero-continuation upper bounds

$$\bar{q}_{ijc}^D = \zeta^D \log \left(1 + e^{\mu_{ijc}^D / \zeta^D} \right), \quad \bar{q}_{ijc}^P = \zeta^P \log \left(1 + e^{\mu_{ijc}^P / \zeta^P} \right).$$

Lemma 1 (Fixed-exposure contraction). *Suppose that both λ^{self} and λ^{agency} are fixed matrices*

that do not vary with α or with the fixed-point iterate. Let

$$\kappa_\Phi = \max \left\{ \max_i \frac{\sum_{j,c} \lambda_{ij}^c \left(1 + \frac{\bar{q}_{ijc}^D}{4\zeta^P} \right)}{r + \delta_i^a}, \right. \\ \left. \max_j \frac{\sum_{i,c} K_i \lambda_{ij}^c \left(1 + \frac{\bar{q}_{ijc}^P}{4\zeta^D} \right)}{r + w_j} \right\}.$$

If $\kappa_\Phi < 1$, then Φ_K is a contraction under the sup norm and the stationary-renewal equilibrium is unique.

Proof. From the closed forms,

$$\left| \frac{\partial \psi_{ijc}^D}{\partial \alpha_i^D} \right| = P_{jic}^P P_{ijc}^D \leq 1.$$

The derivative of a logistic acceptance probability is bounded in absolute value by $1/(4\zeta)$, while $q_{ijc}^D(\alpha) \leq \bar{q}_{ijc}^D$ for $\alpha \geq 0$. Hence

$$\left| \frac{\partial \psi_{ijc}^D}{\partial \alpha_j^P} \right| \leq \frac{\bar{q}_{ijc}^D}{4\zeta^P}.$$

The symmetric bounds hold for ψ_{ijc}^P . It follows that every absolute row sum of the Jacobian of Φ_K is no greater than κ_Φ . The mean-value inequality therefore gives $\|\Phi_K(\alpha) - \Phi_K(\alpha')\|_\infty \leq \kappa_\Phi \|\alpha - \alpha'\|_\infty$, and Banach's theorem applies. $\square$

This condition requires discounting and exit to dominate arrival-weighted surplus. It is sufficient, not necessary, and applies when the route-rate matrices are fixed. It therefore applies to the fixed-route full-replacement map, but it does not establish uniqueness of the behavioral exposure map used in estimation or of the search-retained hybrid map. Reported long-run results instead condition on the maintained numerical equilibrium selection and are labelled best found rather than optimal.

A.5 Continuous Time and Route Accounting

Independent continuous-time Poisson clocks imply that the probability of a self- and an agency-initiated approach arriving at exactly the same instant is zero. This justifies adding the two arrival rates. It does not justify pooling their post-exposure outcomes. When acceptance depends on the initiating route, the fixed point must retain

$$\sum_c \lambda_{ij}^c p_{ijc}, \quad \sum_c \lambda_{ij}^c \psi_{ijc}^D, \quad \text{and} \quad \sum_c K_i \lambda_{ij}^c \psi_{ijc}^P$$

as route-weighted sums. The estimator and policy solver use this accounting throughout.

The stationary-renewal assumption enters the structural continuation values and all long-run policy levels. The raw route-specific acceptance patterns do not require it. The short-run

Table B.1. Sample, Route, and History Construction

Object	Count	Positive cells	Share (%)
<i>A. Sample flow</i>			
Raw December doctor–post cells	2,768,872		
Timeline-invalid cells excluded	38,488		
Final doctor–post cells	2,730,384		
Active doctors	1,132		
Final posts	2,412		
Timeline-invalid posts excluded	34		
<i>B. Route-specific approaches entering the MPEC likelihood</i>			
Sorting (doctor initiated)	2,503	2,481	0.091
Screening (agent initiated)	1,013	1,009	0.037
<i>C. Prior-relationship support</i>			
All final cells	2,730,384	121,065	4.43
Cells with any panel-recorded approach	3,518	1,758	49.97
Cells with a recorded bilateral-status decision	3,381	1,716	50.75
Route-attributable acceptance-likelihood cells	3,281	1,685	51.36

Notes: The raw data contain 2,504 doctor-initiated events. The estimation sample excludes one audited out-of-window Web event, recorded on November 18, 2024 after the post’s half-open observation window ended on November 15. Consequently, 2,503 doctor-initiated events enter the MPEC likelihood. The same exclusion removes one positive sorting cell and one doctor- and post-side sorting acceptance success/trial. Panel C’s first three rows describe the raw final panel; its last row and Panel B describe the post-policy likelihood sample. Repeated in-window approaches remain distinct exposure events. Histories exclude the current post episode and use only information available before its order date.

objective does not re-solve the stationary system after a policy change, but it still inherits the baseline continuation values from the baseline MPEC estimate. Accordingly, neither the short-run nor the long-run exercise should be read as a literal transition forecast under nonstationary entry, seasonal composition, or permanent depletion of matched types.

B Additional Data and Estimation Material

B.1 Episode-Aware Route and Decision Construction

This subsection reports the post-episode selection rule, treatment of reused post identifiers, raw-media route mapping, bilateral status mapping, and the acceptance exclusions described in Section 2.2.

Table B.2. Predetermined Doctor–Facility Relationship Variables

Variable	N	Mean	SD	P25	Median	P75	P95	Max	Nonzero (%)
Prior approaches (365 days)	2,730,384	0.11	1.92	0.00	0.00	0.00	0.00	159.00	2.38
Any prior contract	2,730,384	0.02	0.14	0.00	0.00	0.00	0.00	1.00	2.00
Days since last approach	2,730,384	19.58	127.11	0.00	0.00	0.00	0.00	1,899.33	4.43
Prior agency share (365 days)	2,730,384	0.00	0.07	0.00	0.00	0.00	0.00	1.00	0.66
No prior relationship	2,730,384	0.96	0.21	1.00	1.00	1.00	1.00	1.00	95.57

Notes: All entries are raw, pre-standardization values on the final doctor–post panel. Approach counts and agency shares use the preceding 365 days. Contract, recency, and no-history measures use available records strictly before the post order date. The variables proxy persistent familiarity or match quality but are neither pair fixed effects nor instruments. Days since last approach and prior agency share are coded zero when no prior relationship is observed; the separate no-prior-relationship indicator records that boundary.

B.2 Relationship-History Variables

The history table reports the five main regressors, their information cutoff, window, support, mean, standard deviation, and selected quantiles. Contract history is dated by the recorded contract date; approach counts, route shares, and recency use approach timestamps. No outcome resolved after the current post’s order date enters the main feature set.

B.3 MPEC Implementation and Convergence Diagnostics

Let $\ell(\theta, \alpha)$ denote the sum of the two route-specific Poisson approach-count components and the side-specific Bernoulli trials supplied by route-attributable doctor–post cells, and let $c(\theta, \alpha) = \alpha - \Phi(\theta, \alpha)$. For numerical evaluation, the implementation uses the algebraically equivalent update

$$\begin{aligned}\tilde{\Phi}_i^D &= \frac{\sum_{j,c} \lambda_{ij}^c \{\psi_{ijc}^D + \alpha_i^D p_{ijc}\}}{r + \delta_i^a + h_i^D}, & h_i^D &= \sum_{j,c} \lambda_{ij}^c p_{ijc}, \\ \tilde{\Phi}_j^P &= \frac{\sum_{i,c} K_i \lambda_{ij}^c \{\psi_{ijc}^P + \alpha_j^P p_{ijc}\}}{r + w_j + h_j^P}, & h_j^P &= \sum_{i,c} K_i \lambda_{ij}^c p_{ijc}.\end{aligned}$$

For either side k , with the corresponding exit rate e_k and match hazard h_k , the residuals obey the pointwise identity

$$(r + e_k + h_k) \{\alpha_k - \tilde{\Phi}_k(\theta, \alpha)\} = (r + e_k) \{\alpha_k - \Phi_k(\theta, \alpha)\}.$$

The two residual maps therefore have exactly the same zero set. Writing $\tilde{c}(\theta, \alpha) = \alpha - \tilde{\Phi}(\theta, \alpha)$, the implemented MPEC problem is

$$\max_{\theta, \alpha} \ell(\theta, \alpha) \quad \text{subject to} \quad \tilde{c}(\theta, \alpha) = 0,$$

which is equivalent to imposing $c(\theta, \alpha) = 0$ in the model. The active parameter space contains 92 structural coordinates, including both route contact-rate constants, and 3,544 continuation

values (1,132 doctor values and 2,412 post values). Fourteen direct coordinates with exact separation or no support are fixed at zero. The calculation does not profile the route constants or impose the fixed- α likelihood score as an additional restriction. The exposure scales are normalized to $\zeta^{\text{self}} = \zeta^{\text{agency}} = 1$; neither an inverse exposure scale nor its logarithm is an estimated coordinate. Consequently the same continuation values used in acceptance enter the two exposure equations with coefficient minus one.

The reported MPEC estimate has a six-block unprofiled log likelihood of $-35,423.373079$. The sup norm of the raw equilibrium residual is 9.855422×10^{-7} , and the solver-scale unprojected KKT sup norm is 0.188066645. Because the solver divides the total negative log likelihood by 10,031, the corresponding raw-total unprojected KKT sup norm is 1,886.4965. The equilibrium residual is below the reported strict feasibility threshold of 10^{-6} , whereas the stationarity diagnostic does not pass the preferred or strict first-order thresholds reported in Table B.8. These diagnostics therefore do not certify an exact MPEC solution, a local first-order solution, constraint qualification, second-order sufficiency, a local or global optimum, or equilibrium uniqueness. A joint MPEC covariance matrix has not been computed, so the exhibits contain no standard errors, p -values, or significance stars.

B.4 Covariate Coefficient Estimates

The full tables report the 90 active covariate slopes in their exact implementation order. The two route-specific acceptance constants remain in the structural vector but are not reported because their levels are not separately interpreted. Fourteen unsupported coordinates in the corresponding full vectors are held at zero and are not interpreted as estimated zeros. The exposure loadings on continuation values are fixed normalizations rather than estimated coefficients.

Table B.3. Full MPEC Point Coefficients: Doctor acceptance

Active margin coordinate	Point estimate	theta92 position
const	-0.3078	0
log_distance_km	-0.2196	1
age	0.1246	2
experience	0.1530	3
hours	-0.0190	4
log_salary_high	0.0319	5
pref_match_dummy	0.1673	6
day_match_dummy	0.2222	7
spec_match_dummy	0.1462	8
gairai_match_dummy	0.1290	9
houmon_match_dummy	0.0042	10
byoutou_match_dummy	0.0933	11
touseki_match_dummy	0.0253	12
ningen_doc_match_dummy	0.0335	13
naishikyou_match_dummy	0.0193	14
oushin_match_dummy	-0.0014	15
jiyuu_match_dummy	0.0422	16
rel_prior_approach_count_12m	0.0296	17
rel_prior_contract_dummy	0.0023	18
rel_days_since_last_approach	0.0024	19
rel_prior_agency_share_12m	0.0094	20
rel_no_prior_history_dummy	0.0039	21

Notes: The table follows the exact theta92 name order and reports active point coordinates only. Fourteen unsupported full-vector coordinates are held at zero and are not displayed. Exposure continuation-value loadings are normalized to one and are not estimated rows. No uncertainty measures or significance symbols are supplied.

Table B.4. Full MPEC Point Coefficients: Post acceptance

Active margin coordinate	Point estimate	theta92 position
const	0.3587	23
log_distance_km	-0.2073	24
age	-0.0410	25
experience	-0.0105	26
hours	0.1658	27
log_salary_high	0.0166	28
pref_match_dummy	0.1183	29
day_match_dummy	0.0934	30
spec_match_dummy	0.1000	31
gairai_match_dummy	0.0246	32
houmon_match_dummy	-0.0074	33
byoutou_match_dummy	0.0677	34
touseki_match_dummy	0.0118	35
ningen_doc_match_dummy	-0.0094	36
naishikyou_match_dummy	0.0119	37
oushin_match_dummy	0.0014	38
doukei_match_dummy	0.0001	39
jiyuu_match_dummy	0.0200	40
rel_prior_approach_count_12m	0.0295	41
rel_prior_contract_dummy	0.0034	42
rel_days_since_last_approach	0.0023	43
rel_prior_agency_share_12m	0.0199	44
rel_no_prior_history_dummy	0.0014	45

Notes: The table follows the exact theta92 name order and reports active point coordinates only. Fourteen unsupported full-vector coordinates are held at zero and are not displayed. Exposure continuation-value loadings are normalized to one and are not estimated rows. No uncertainty measures or significance symbols are supplied.

Table B.5. Full MPEC Point Coefficients: Sorting exposure

Active margin coordinate	Point estimate	theta92 position
const	-7.6582	47
log_distance_km	-1.0208	48
age	0.3801	49
experience	0.6709	50
hours	-0.0408	51
log_salary_high	0.2874	52
pref_match_dummy	0.3055	53
day_match_dummy	0.6231	54
spec_match_dummy	0.5500	55
gairai_match_dummy	0.2918	56
houmon_match_dummy	-0.0404	57
byoutou_match_dummy	0.2967	58
touseki_match_dummy	0.1709	59
ningen_doc_match_dummy	0.0553	60
naishikyou_match_dummy	0.1319	61
oushin_match_dummy	-0.0307	62
doukei_match_dummy	0.0422	63
jiyuu_match_dummy	0.2557	64
rel_prior_approach_count_12m	0.2099	65
rel_prior_contract_dummy	0.0795	66
rel_days_since_last_approach	0.1014	67
rel_prior_agency_share_12m	0.1015	68
rel_no_prior_history_dummy	0.0210	69

Notes: The table follows the exact theta92 name order and reports active point coordinates only. Fourteen unsupported full-vector coordinates are held at zero and are not displayed. Exposure continuation-value loadings are normalized to one and are not estimated rows. No uncertainty measures or significance symbols are supplied.

Table B.6. Full MPEC Point Coefficients: Screening exposure

Active margin coordinate	Point estimate	theta92 position
const	-10.6129	70
log_distance_km	-1.2786	71
age	-0.2581	72
experience	-0.0017	73
hours	1.2391	74
log_salary_high	0.1650	75
pref_match_dummy	0.2404	76
day_match_dummy	-0.0581	77
spec_match_dummy	0.2658	78
gairai_match_dummy	-0.3827	79
houmon_match_dummy	-0.1011	80
byoutou_match_dummy	0.1175	81
touseki_match_dummy	0.0798	82
ningen_doc_match_dummy	-0.2127	83
naishikyou_match_dummy	0.0746	84
oushin_match_dummy	-0.0034	85
jyuu_match_dummy	0.1643	86
rel_prior_approach_count_12m	0.1189	87
rel_prior_contract_dummy	0.0803	88
rel_days_since_last_approach	0.0496	89
rel_prior_agency_share_12m	0.4105	90
rel_no_prior_history_dummy	-0.0061	91

Notes: The table follows the exact theta92 name order and reports active point coordinates only. Fourteen unsupported full-vector coordinates are held at zero and are not displayed. Exposure continuation-value loadings are normalized to one and are not estimated rows. No uncertainty measures or significance symbols are supplied.

Table B.7. Full Point Distance-Equivalent Transformations

Variable	Sorting	Screening	Doctor acceptance	Post acceptance
Doctor age	-28.5	19.9	-40.0	19.4
Experience	-44.6	0.1	-46.6	4.7
Hours	3.7	-58.2	8.1	-51.3
Posted salary	-22.4	-11.0	-12.3	-7.0
Preference match	-23.6	-15.6	-49.6	-40.1
Day match	-42.3	4.2	-59.8	-33.3
Specialty match	-38.4	-17.1	-45.0	-35.2
Outpatient match	-22.7	30.9	-41.0	-10.1
Emergency match	—	—	—	—
Home-visit match	3.6	7.4	-1.7	3.3
Ward match	-23.0	-7.9	-31.8	-25.4
Dialysis match	-14.0	-5.5	-9.8	-5.0
Health-check match	-4.8	16.1	-12.8	4.2
Endoscopy match	-11.0	-5.1	-7.6	-5.0
Surgery match	—	—	—	—
House-call match	2.7	0.2	0.6	-0.6
Work-with-peer match	-3.7	—	—	-0.1
Flexible-work match	-20.2	-10.9	-15.9	-8.3
Occupational-health match	—	—	—	—
Prior approaches (past 365 days)	-16.9	-8.0	-11.4	-12.0
Prior contract	-6.8	-5.5	-0.9	-1.5
Days since last approach	-8.6	-3.4	-1.0	-1.0
Prior agency share (past 365 days)	-8.6	-25.1	-3.8	-8.3
No prior relationship	-1.8	0.4	-1.6	-0.6

Notes: Entries are percentage changes in distance for a one-sample-SD continuous change or a zero-to-one indicator change. Dashes mark an unsupported numerator coordinate. The unit exposure-loading normalization cancels from each within-margin ratio. No uncertainty measures or significance symbols are supplied.

B.5 Numerical Diagnostics

Table B.8. MPEC Point-Estimate and Fit Diagnostics

Diagnostic	Realized value	Gate / interpretation
Reported estimate	MPEC	Audited coefficient and continuation-value point
Doctor-post cells	2,730,384	Exact materialized schema
Doctors / posts	1,132 / 2,412	Exact ID order
Active / held full coordinates	92 / 14	Exact theta92 projection / zero complement
Six-block unprofiled log likelihood	-35423.373079	All six blocks independently replayed
Implemented/Form-A raw equilibrium residual (sup norm)	9.855422e-07	Independently replayed
Solver-scale unprojected KKT sup norm	1.880666e-01	Independent post-result refit
Raw-total unprojected KKT sup norm	1886.4965	Total-objective scale
Reported acceptance rule	$\ \tilde{c}\ _\infty \leq 10^{-5}$ and scaled KKT ≤ 0.25	PASS
Strict equilibrium-feasibility rule	$\ \tilde{c}\ _\infty \leq 10^{-6}$	PASS
Strict first-order rule	Scaled KKT ≤ 0.001	NOT PASSED
Preferred first-order rule	Scaled KKT ≤ 0.05	NOT PASSED
Direct/legacy predictor and primitive parity (sup)	0.000e+00	PASS
Six-block likelihood bridge parity (sup)	0.000e+00	PASS
Form-A and baseline-policy map parity (sup)	1.776e-15	PASS
Observed / fitted contacts, sorting	2,503 / 2503.000	Volume-fit diagnostic
Observed / fitted contacts, screening	1,013 / 1013.000	Volume-fit diagnostic
Raw-sample log-distance SD	0.899556299578	SHA-bound point transform
Exact / local first-order / CQ / SOSC certificate	No / No / No / No	Not certified
Local / global optimum / equilibrium uniqueness certificate	No / No / No	Not certified
Joint MPEC inference	Not computed	No SEs, p-values, or significance symbols

Notes: The reported MPEC estimate satisfies the reported numerical acceptance rule and the strict equilibrium-feasibility rule, but not the strict or preferred first-order rules. The implemented/ Form-A raw equilibrium residual is 9.855422e-07; the solver-scale and raw-total unprojected KKT sup norms are 1.880666e-01 and 1886.4965. The latter uses total NLL, while the solver divides by 10,031. These diagnostics do not certify an exact MPEC solution, local first-order solution, CQ, SOSC, a local or global optimum, or equilibrium uniqueness. Joint MPEC covariance is not computed, so the table reports no standard errors, p-values, or significance symbols.

C Additional Policy Results

Figures C.1 and C.2 display the access–value comparisons used to assess whether doctor search should remain available. Figures C.3 and C.5 then report the recommendation- volume decomposition and market-expansion comparison with doctor search retained. In the search-retained exercises, every long-run candidate recomputes doctor search from the solved continuation values; under full replacement in Figure C.2, doctor search is zero by construction. Table C.1 reports the equilibrium residuals and stopping outcomes for the best-found rules used in these comparisons.

Table C.1. Diagnostics for the Reported Best-Found Policy Searches

Intervention	Budget search	Start lineage	Accepted	Stop reason	Max residual	Max row error	Max agency-cap violation	Realized-total overage	Self-search error	Eq. tol.
Search retained	Current agency budgets	Short-run-best start	8	converged below tolerance	9.81e-06	2.59e-11	7.11e-15	9.03e-01	0.00e+00	1e-05
Search retained	Doubled equal agency budgets	Equal-budget random start	7	max iter cap	9.44e-06	1.00e-11	1.08e-19	1.49e+00	0.00e+00	1e-05
Full replacement	Current total budgets	Current policy profile	8	max iter cap	8.64e-06	1.81e-10	7.11e-15	7.11e-15	—	1e-05

Notes: Row error and agency cap refer to the platform-controlled agency matrix. For search retained, realized-total overage is reported for transparency and is not a policy-feasibility gate. Self-search error checks saved doctor-initiated counts against the behavioral search equation. The start-lineage column distinguishes a design imported from the prior estimate from any continuation values, coefficients, or state: only the policy profile is imported and it is re-evaluated under the current estimate. These diagnostics support best-found, maintained-branch claims rather than global optimality.



Figure C.1. Access and value when agent screening is redesigned and doctor search responds in equilibrium.



Figure C.2. Access and value when doctor search is removed and all designed contacts use the agent route. The current hybrid is outside the candidate class.

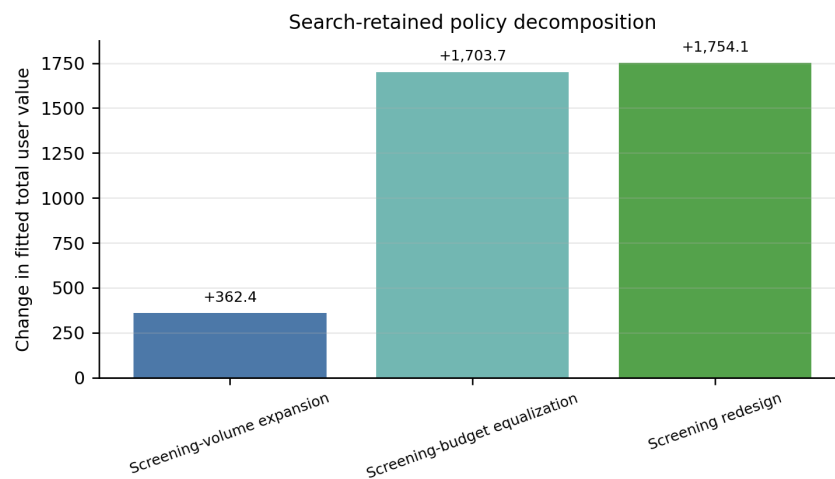


Figure C.3. Value changes from expanding and equalizing controlled screening exposure and then redesigning pair allocations while doctor search responds.

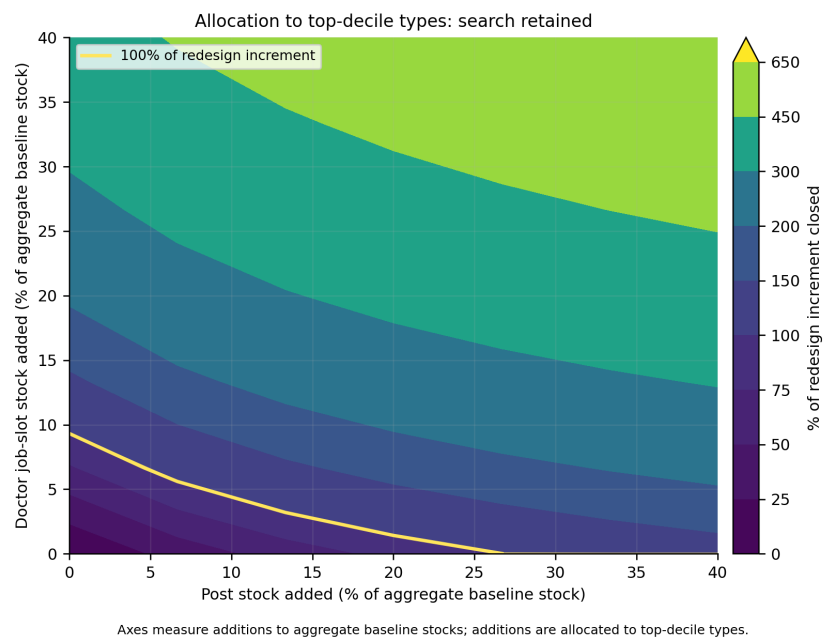


Figure C.5. Market-composition equivalents of the equal-budget redesign increment with doctor search retained.